

DAQ Validator Operator Manual

Software workflow for pre-run DAQ validation using DC and sine stimulus runs

Purpose. This manual explains how to run the current DAQ validation workflow, how the UI displays results, and how the scoring layer interprets the two required test runs: a DC/short run and a 1 kHz sine run.

Scope. The workflow is intended for pre-run validation of the measurement chain below the sensor interface, so an operator can quickly identify suspicious channels before engine testing.

1. System overview

The current stack has four parts: LabVIEW-based acquisition, a Python scoring layer, a simple Tkinter operator UI, and a manifest-driven data bundle.

- LabVIEW acquires the data from the NI PXI-4472 and can also perform device self-test and self-calibration before the run. NI states that self-calibration uses an onboard reference and no external signal connections are necessary, while self-test verifies basic I/O and may require exclusive device access [2][3].
- The PXI-4472 is an 8-channel, simultaneously sampled DSA module with pseudodifferential inputs, AC or DC software-selectable coupling, 24-bit delta-sigma ADCs, built-in antialiasing filters, and a 102.4 kS/s maximum sample rate [1][4].
- Python reads the acquired files, computes the channel scores, and writes the result file used by the UI. The NI-DAQmx Python package supports TDMS logging, which is a useful production option when you move beyond CSV [5].
- The UI shows only operator-relevant outputs: channel, verdict, scores, spectral metrics, and reasons.

2. Files in the working directory

File	Role	Required?
simple_ui.py	Operator interface that launches scoring_layer.py and displays the CSV results	Yes
scoring_layer.py	Main analysis and scoring code	Yes
manifest.json	Points to the DC and sine data files and defines expected values	Yes
dc_run.csv	DC or shorted-input acquisition	Yes
sine_run.csv	1 kHz sine acquisition	Yes
daq_validation.csv/.json	Generated output from the scoring layer	Created by software

Important. The UI expects the result CSV to be in the same directory as the scripts. The manifest uses relative file names, so it is easiest to run everything from one working folder.

3. Operator procedure

1. Allow the PXI-4472 and the chassis to warm up. NI specifies a 15-minute warm-up to achieve rated accuracy [1].

2. Run the LabVIEW acquisition sequence for the DC mode and the sine mode. The two acquisitions must be separate because they use different excitations.
3. Launch `simple_ui.py`.
4. Click Run Validation.
5. Review the verdicts and reasons. PASS means the channel is acceptable, SUSPECT means it deserves inspection, and FAIL means it should be treated as bad until cleared.

4. Test sequence used by the software

The scoring layer combines two measurements per channel. This is deliberate: the DC run is good at exposing offset, flatline, and noise problems, while the sine run is good at exposing gain, phase, clipping, distortion, and spurious tones.

- DC/short run: the input is shorted or driven to a known DC level. This checks offset, noise floor, flatline behavior, saturation, and channel-to-channel consistency.
- Sine run: the input is driven with a 1 kHz sine wave. The NI 447x specifications themselves use a 1 kHz, -1 dBFS sine-wave condition for SFDR and THD reporting, with 131,072 FFT samples in the published plots [1].
- The sine run should remain stable for the full record length. The current analyzer uses the waveform data as a single contiguous block per channel.

5. How the scoring works

The software gives each channel a score from 0 to 100 for each mode, then combines the two mode scores into one final verdict. The exact scoring is transparent and rule-based, not machine learning.

5.1 DC-mode metrics

- Mean offset: difference between the measured mean and the expected DC level from the manifest.
- Standard deviation and RMS: used as a noise and stability measure.
- Peak-to-peak value: useful for identifying gross instability.
- Clipping fraction: fraction of samples near full-scale; if this is large, the channel is likely saturated.
- Flatline detection: very small standard deviation indicates a dead path, frozen data, or a disconnected channel.
- Robust outlier check: the channel is compared with the group median and median absolute deviation (MAD) so that one channel can be flagged even if the absolute values are not yet out of spec.

5.2 Sine-mode metrics

- Sine fit amplitude: a least-squares fit extracts the fundamental amplitude for the 1 kHz tone.
- Sine fit phase: used to check timing alignment between channels.
- SINAD: signal-to-noise-and-distortion ratio, estimated from the FFT.
- THD: total harmonic distortion, estimated from the FFT.
- SFDR: spurious-free dynamic range, estimated from the FFT.
- Clipping fraction and flatline checks are also applied in sine mode.

5.3 Combined score and verdict

The current code combines the two mode scores as: $\text{combined} = 0.4 \times \text{DC score} + 0.6 \times \text{sine score}$.

- PASS: both mode scores are low and the combined score is low.
- SUSPECT: moderate deviation in one or both modes, or the combined score is elevated.

- FAIL: hard fault conditions, flatline, clipping, NaN/Inf, or very large mode scores.

In the current implementation, hard faults override everything else. For example, a flatline DC record or a heavily clipped sine record is treated as FAIL even if some other metrics look acceptable.

5.4 Why the code uses two acquisitions

- A single waveform cannot exercise all the checks at once because the DC test and sine test measure different failure modes.
- Separate acquisitions keep the math simpler and the operator workflow clearer.
- This also makes the data easier to archive and trend over time.
- For deeper characterization, additional stimulus modes can be added later without changing the pre-run gate.

6. What the UI shows

- Channel: the channel name taken from the CSV header and manifest.
- Verdict: PASS, SUSPECT, or FAIL.
- Combined score: the final weighted score used for sorting and prioritization.
- DC score: score from the DC/short run.
- Sine score: score from the 1 kHz sine run.
- SINAD, THD, SFDR: the spectral metrics derived from the sine run.
- Reasons panel: human-readable explanation of why the channel was flagged.

Row colors are mapped as follows: green = PASS, yellow = SUSPECT, red = FAIL.

7. What data must be acquired

- A DC or shorted-input record for every channel in the test set.
- A 1 kHz sine record for the same channels, sampled with the same sample rate and record length.
- The two records must be separate files so the analyzer can apply the correct metric set.
- The manifest must identify the files and the expected values for each mode.
- The current analyzer expects each CSV file to have the channel names in the first row and one sample per row after that.

Recommended record size. The current configuration uses 131,072 samples per record, which is also the FFT size used in the NI 447x specification plots for SFDR and THD [1].

8. How to interpret results

Verdict	Meaning	Operator action
PASS	Channel is consistent with the expected behavior	No action needed
SUSPECT	Channel is unusual and should be inspected	Check wiring, connector, sensor path, and rerun
FAIL	Channel shows a strong fault indication	Do not trust the channel until the issue is found

Practical rule. When several channels are flagged, start with the worst combined scores first. The software is designed to narrow the operator's search from thousands of channels down to a short suspect list.

9. Troubleshooting

- If the analyzer fails immediately, verify that `scoring_layer.py`, `manifest.json`, `dc_run.csv`, and `sine_run.csv` are in the same folder as the UI.
- If no channels appear, check that the CSV header row contains the channel names and that both files have the same channel order.
- If many channels are marked FAIL, first check the excitation setup, grounding, and whether the correct stimulus was applied to the correct file.
- If the sine metrics look poor across every channel, verify that the sine source frequency is 1 kHz and that the acquisition sample rate and record length are correct.
- For calibration-sensitive work, re-run self-test and self-calibration after warm-up and after large temperature changes [2][3].

10. Notes on deeper characterization

The current manual describes the fast pre-run validator. For periodic engineering audits, you can add more stimulus modes later, such as square, step, impulse, and histogram/code-density tests. ADC literature treats histogram tests as a standard way to estimate DNL and INL, but it also notes that they require many samples and a very well-defined stimulus. FFT-based sine testing is likewise a standard dynamic test and is commonly paired with histogram testing in a full ADC test plan [6][7].

References

- [1] NI 447x Specifications. National Instruments. Covers the PXI/PCI-4472 family, including 8 simultaneously sampled channels, pseudodifferential input, AC/DC coupling, 24-bit delta-sigma ADCs, 102.4 kS/s maximum sample rate, warm-up time, and published SFDR/THD test conditions.
- [2] NI System Configuration API Reference: Self-Test and Self-Calibrate. National Instruments. Describes self-test and self-calibration behavior, including the onboard reference and the need to disconnect AC input signals before self-calibration.
- [3] NI-DAQmx LabVIEW API Reference: DAQmx Self Test Device VI and DAQmx Self Calibrate VI. National Instruments. Documents the device self-test and self-calibration VIs.
- [4] PXI-4472 product page. National Instruments. Summarizes the PXI-4472 as a dynamic signal acquisition module with built-in antialiasing filters and simultaneous digitization.
- [5] NI-DAQmx Python Documentation. NI-supported Python wrapper. Documents TDMS logging for high-speed acquisition and logging to disk.
- [6] Analog Devices, Data Conversion Handbook, Chapter 5: Testing Data Converters. Describes histogram/code-density testing, DNL, INL, and the relationship between histogram and FFT methods.
- [7] Analog Devices AN-835: Understanding High Speed ADC Testing and Evaluation. Discusses histogram tests, sample counts, and sine-wave curve-fit methods for ADC evaluation.